



Effect of seed pelleting on seed quality and seedling growth of small seed of maize

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Abstract

Background and Objective: Small-sized maize seeds are often less preferred by farmers because of their lower vigor and poorer field establishment, resulting in reduced utilization and economic losses. This study evaluated the effectiveness of different pelleting materials in improving the physical and physiological quality of small seeds of maize (*Zea mays* L.) inbred line Tak Fa 1.

Methodology: The experiment was conducted at the Khon Kaen Seed Research and Development Center and the Nakhon Sawan Field Crops Research Center from May 2023 to July 2024. Maize seeds were classified into small (16/64 inch), medium (18/64 inch), and large (20/64 inch) size classes. Small seeds were pelleted using hydroxypropyl methylcellulose as a binder with bentonite, vermiculite, talcum, or pumice as filler materials. Unpelleted small, medium, and large seeds served as controls. Laboratory experiments were arranged in a completely randomized design, while field experiments followed a randomized complete block design with four replications. Seed physical traits, germination, vigor, field emergence, and seedling growth were evaluated.

Main Results: Pelleting increased seed size from 16/64 inch to approximately 18/64 inch and improved seed uniformity. Germination remained high under both laboratory and field conditions, indicating that pelleting did not adversely affect seed viability. Bentonite- and vermiculite-pelleted seeds exhibited significantly greater vigor than unpelleted small seeds, as evidenced by higher germination after

accelerated aging and faster emergence under field conditions. These materials also improved seedling establishment and overall field performance. Talcum and pumice promoted early seedling growth but were less effective in maintaining seed vigor under aging stress.

Conclusions: Bentonite and vermiculite were the most effective pelleting materials for improving seed size, vigor, stress tolerance, and field establishment of small Tak Fa 1 maize seeds. Seed pelleting enhanced the value and utilization of small seeds without compromising seed quality, resulting in performance comparable to commercially accepted seed grades.

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INTRODUCTION

High-quality seed is essential for reliable crop establishment and can increase yield by up to 30% (Afzal, 2023). Seed quality, however, declines during storage, resulting in reduced germination and vigor. Seed enhancement, consisting of seed priming and pelleting, is a post-harvest seed treatment that improves germination and seedling growth. Although these technologies cannot reverse physiological deterioration, they can optimize germination conditions and enhance seed handling characteristics, especially when pelleting is applied (Taylor *et al.*, 1998; Siri, 2015). Pelleting is commonly used for small, high-value seeds to modify shape and size, incorporate beneficial materials, and improve compatibility with mechanical planters (Pedrini *et al.*, 2017; Afzal *et al.*, 2020; Taylor, 2020).

Previous studies have demonstrated that seed pelleting can increase seed size and weight, improve uniformity, and enhance planter performance in crops such as lettuce, carrot, onion, sugar beet, and maize (Smith and Miller, 1987; Taylor *et al.*, 1998; Pedrini *et al.*, 2017). In maize, pelleting has been

shown to improve handling and planting efficiency of small seeds and, in some cases, enhance early seedling vigor through the delivery of nutrients or other beneficial compounds (Sakhamula and Siri, 2012). Recent research also shows that seed pelleting can improve germination, seedling growth, and uniformity in tomato, lettuce, and other horticultural crops, highlighting the importance of selecting suitable filler and binder materials (Javed and Afzal, 2018; Jeephet *et al.*, 2022; 2024). These studies indicate that pelleting provides both physical benefits (increased size and uniformity) and potential biological enhancements (improved vigor), although positive effects on germination and seedling growth are not always consistent and may depend on genotype and pelleting materials (Afzal *et al.*, 2020).

In Thailand, the inbred maize line Tak Fa 1 represents an important parental line for Thailand's hybrid maize as the certified hybrid cultivars Nakhon Sawan 2, Nakhon Sawan 3, and Nakhon Sawan 4, released in 2006, 2009, and 2019, respectively. Tak Fa 1 was developed from the Pioneer 3003 F₂ population during 2000–2003, with selection focusing

on hybrid performance, downy mildew resistance, and drought tolerance (Nakhon Sawan Field Crop Research Center, 2021). After harvest and seed processing, maize seeds are typically separated into small, medium, and large size classes, each showing distinct physiological and agronomic performance.

Seed size is a critical determinant of seed quality, seedling vigor, and yield potential. Larger seeds generally contain higher storage reserves, contributing to enhanced stress tolerance, uniform emergence, and superior field performance (Elmore and Abendroth, 2005; Ambika *et al.*, 2014). Conversely, small seeds often produce weaker seedlings, exhibit lower mechanical planting efficiency, and may fail to meet commercial planting standards (Sakhamula and Siri, 2012; Wongkupthai *et al.*, 2018). Despite lacking inherent physiological disadvantages, small maize seeds are routinely rejected by farmers due to poor compatibility with mechanical planters, especially for the Tak Fa 1 inbred line. In Tak Fa district, Nakhon Sawan province, farmers commonly avoid size-16 seeds (16/64 inch) in favor of size-18 and size-20 seeds, resulting in the diversion of small seeds to livestock feed and economic losses estimated at 140 million baht annually. Seed rejection reduces plantable seed availability and seed-use efficiency, and has broader implications for food and feed security (Wongkupthai *et al.*, 2018).

Seed pelleting offers a practical solution to these challenges by increasing small seed size to medium, enhancing uniformity, and improving planter compatibility. Pelleting can also serve as a carrier for nutrients, growth-promoting bacteria, or protective agents to support germination and seedling growth (Sakhamula and Siri, 2012; Jeepphet *et al.*, 2024). The success of pelleting depends on selecting filler materials that are inexpensive, readily available, water-absorbent, compatible with binders, and capable of cracking or dissolving during germination. Common

mineral-based fillers such as bentonite, vermiculite, talc, and pumice are widely used due to their favorable physical and chemical properties (Smith and Miller, 1987; Afzal, 2023). Therefore, the objective of this study was to identify suitable pelleting materials for small-sized maize seeds and to evaluate their effects on seed physical traits, germination, and seedling growth. The ultimate goal was to improve seed-size management, enhance the economic value of small seeds, and increase their utilization in commercial maize production.

Despite extensive research demonstrating the benefits of seed pelleting in improving seed handling, size uniformity, and planter performance, most studies have focused on vegetable crops and commercial hybrid seeds (Taylor *et al.*, 1998; Pedrini *et al.*, 2017; Afzal *et al.*, 2020). Previous reports have shown that pelleting materials can influence germination and seedling growth through their physical and chemical properties, including water absorption, aeration, and nutrient retention (Taylor *et al.*, 1998; Javed and Afzal, 2018; Jeepphet *et al.*, 2024). However, the effects of different pelleting materials on small-sized seeds of maize inbred lines remain poorly understood. In particular, there is limited information regarding whether pelleting can effectively upgrade small maize seeds to commercially acceptable sizes without adversely affecting germination and seedling establishment. Furthermore, comparative evaluations of commonly available mineral-based pelleting materials for improving the utilization value of small seeds in maize breeding and seed production systems are lacking.

The maize inbred line Tak Fa 1 represents an important parental line in Thailand's hybrid maize seed industry. However, small-sized seeds are frequently rejected by farmers because of poor compatibility with mechanical planting equipment, resulting in substantial economic losses. Identifying suitable pelleting materials that can improve seed

size while maintaining high physiological quality could increase the commercial value and utilization of these otherwise discarded seeds.

Therefore, the objectives of this study were (1) to evaluate the effectiveness of different pelleting materials in increasing the size and weight of small maize seeds, and (2) to determine their effects on seed germination and seedling growth. We hypothesized that (1) pelleting would increase the physical dimensions and weight of small maize seeds to levels comparable with larger commercial seed grades without reducing germination performance, and (2) the magnitude of these effects would differ among pelleting materials because of their contrasting physical properties, particularly water-holding capacity, porosity, and interaction with the binder, which influence seed imbibition and early seedling development.

MATERIALS AND METHODS

The maize inbred line Tak Fa 1 was planted in November 2022 and harvested in March 2023 at the Nakhon Sawan Field Crop Research Center (NSFCRC), Nakhon Sawan province, Thailand. After harvest, kernels were manually shelled from the cobs, cleaned, and impurities were discarded prior to seed size classification. Seeds were separated into small, medium, and large size classes using 16/64-, 18/64-, and 20/64-inch sieves, respectively.

All seed lots met national seed quality standards, with a minimum germination percentage of 75% and seed purity of at least 98% (Ministry of Agriculture and Cooperatives, 2013). Seed moisture content at the start of the experiment was approximately 10–11%.

The experiment was conducted from May 2023 to July 2024 at the Nakhon Sawan Field Crop Research Center. Seed Laboratory and the Khon Kaen Seed Research and Development Center. Seven treatments were evaluated as follows: Trt 1 =

small-sized seeds (16/64 inch) pelleted with bentonite, Trt 2 = small-sized seeds (16/64 inch) pelleted with vermiculite, Trt 3 = small-sized seeds (16/64 inch) pelleted with talcum, Trt 4 = small-sized seeds (16/64 inch) pelleted with pumice, Trt 5 = unpelleted small-sized seeds (16/64 inch; S16), Trt 6 = unpelleted medium-sized seeds (18/64 inch; S18), and Trt 7 = unpelleted large-sized seeds (20/64 inch; S20).

Pelleting materials were selected based on their local availability in Thailand and cost-effectiveness. Four filler materials were evaluated: bentonite (Union Chemical Ltd., Bangkok, Thailand), vermiculite (Greeningplus, Bangkok, Thailand), talcum, and pumice (Union Chemical Ltd., Bangkok, Thailand). Vermiculite was pre-processed by grading with a binder solution using a high-speed mixer operating at 20,000 rpm for 45 s (400 g vermiculite, 2 min) to obtain a uniform particle size suitable for pelleting.

Seed pelleting was performed using an SKK08 rotary seed coater operated at 350 rpm. Small-sized maize seeds (16/64 inch) were pelleted using a 0.5% (w/v) hydroxypropyl methylcellulose (HPMC) solution as a binder, applied via a pneumatic spray system. Pelleting materials were applied in successive layers to achieve uniform coating and to increase seed size from 16/64 inch to 18/64 inch. A total of 350 mL of binder solution and 400 g of filler material were used per kilogram of seed. Pelleted seeds exhibited increased size and weight, improved shape uniformity, and a smoother surface texture.

After pelleting, seeds were dried in mesh bags using a KSD120 hot-air dryer at 35°C until seed moisture content returned to pre-pelleting levels (10–11%).

Pelleting Difficulty

Pelleting difficulty was assessed based on variation among filler materials during the coating process. Ease of adhesion and coverage of the seed

surface was rated using a 1–5 ordinal scale, where 1 = very difficult, 2 = difficult, 3 = moderate, 4 = easy, and 5 = very easy, following Buakaew and Siri (2018).

Seed Quality Parameter

Seed quality of pelleted and unpelleted maize seeds was evaluated under laboratory and field conditions using physical, physiological, and vigor-related parameters.

Seed weight

Seed weight was determined by weighing 100 randomly selected seeds per treatment using a precision balance (± 0.01 g). Measurements were conducted with four replications.

Seed size

Seed size was determined by measuring the maximum diameter of 10 randomly selected seeds per treatment, with four replications. Seed diameter was recorded at the widest point using a vernier caliper, and mean seed diameter was calculated for each treatment.

Seed dissolution

Seed dissolution was evaluated by recording the time required for the pelleting material to initiate disintegration when pelleted seeds were immersed in water. Ten seeds were randomly selected per treatment, with four replications, and dissolution time was defined as the point at which the coating began to detach from the seed surface (Anderson *et al.*, 1970).

Seed standard germination

Standard germination tests were conducted according to International Seed Testing Association protocols (ISTA, 2022). For each treatment, 100 seeds were randomly selected and germinated using the between-paper (BP) method with four replications.

Samples were incubated in alternating temperatures of 30°C for 8 h and 20°C for 16 h at 80% relative humidity under 180 μ E light intensity. First and final germination counts were recorded at 4 and 7 days, respectively.

Germination (%) =

$$\frac{\text{Number of germinated seeds}}{\text{Total seeds tested}} \times 100$$

Speed of germination

Speed of germination was calculated based on daily counts of newly germinated seedlings over a 7-day period, following AOSA (1983). Germination speed was expressed as plants per day using the formula:

$$\text{Speed of germination (plants/day)} = \sum(n/d)$$

where n = number of newly germinated seeds and d = number of days after sowing.

Accelerated aging test (AA test)

Seed vigor was further evaluated using an accelerated aging (AA) test. Seeds were exposed to high temperature and relative humidity conditions, followed by standard germination testing according to ISTA (2022). Post-aging germination was also assessed under field conditions to compare laboratory and field responses following stress exposure.

Field germination

Field germination tests were conducted at the Khon Kaen Seed Research and Development Center. The experiment was arranged in a randomized complete block design with four replications. For each treatment, 100 seeds per replication were randomly assigned and sown in soil beds at a depth of 10 mm and covered with a 5-mm layer of sand. Each replication consisted of a separate plot within a soil bed, and treatments were randomly allocated within each block to minimize environmental variation. Mean daily air temperatures during the test period ranged

from 25 to 30°C under natural field conditions. Germination percentage, speed of germination, and seedling growth parameters were evaluated using the same criteria applied in laboratory assessments.

Seedling growth

Seedling vigor was assessed 14 d after sowing under field conditions. From each plot, seedlings were randomly sampled for measurement. Shoot length was measured from the soil surface to the junction of the leaf sheath and blade of the most recently emerged leaf, while root length was measured from the shoot base to the tip of the longest root, following the guidelines of the Plant Variety Protection Division (2005).

Statistical Analysis

All experiments were conducted with four replications. Laboratory experiments were arranged in a completely randomized design (CRD), whereas field experiments were conducted in a randomized complete block design (RCBD). Data were tested for normality prior to analysis and subsequently subjected to analysis of variance (ANOVA) using the general linear model (GLM) procedure in IRRISTAT software package.

For experiments arranged in a CRD, the statistical model was: $y_{ij} = \mu + T_i + \varepsilon_{ij}$ where y_{ij} is the observation from treatment i (seed size, seed weight, seed dissolution, seed germination, speed of germination, AA test, shoot length and root length), μ is the overall mean of each variable, T_i is the treatment effect (i = Trt 1 small-sized seeds (16/64 inch) pelleted with bentonite, Trt 2 small-sized seeds (16/64 inch) pelleted with vermiculite, Trt 3 small-sized seeds (16/64 inch) pelleted with talcum, Trt 4 small-sized seeds (16/64 inch) pelleted with pumice, Trt 5 unpelleted small-sized seeds (16/64 inch; S16), Trt 6 unpelleted medium-sized seeds (18/64 inch; S18), and Trt 7 unpelleted large-sized

seeds (20/64 inch; S20), and ε_{ij} is the experimental error. For experiments arranged in an RCBD, the statistical model was: $y_{ij} = \mu + B_i + T_j + \varepsilon_{ij}$ where y_{ij} is the observation from treatment i (seed germination, speed of germination, AA test, shoot length, and root length), μ is the overall mean of each variable, B_i is the block effect, T_j is the treatment effect (same treatment as CRD), and ε_{ij} is the experimental error. Treatment means were compared using Duncan's multiple range test (DMRT) at the 5% probability level. Differences among means were considered significant at $P < 0.05$. Data are presented as means $\pm$ standard deviation.

RESULTS AND DISCUSSION

Pelleting Difficulty

The level of difficulty related to the type of pelleting material used for seed pelleting formation. The results from maize seed pelleting with four different materials—bentonite, vermiculite, talcum, and pumice—show that vermiculite was very easy to form pellets compared to the other treatments. Particularly, vermiculite provides a lot of flexibility during pelleting. Vermiculite is a natural mineral that expands when heated to around 800°C. It is neutral, lightweight, insoluble in water, and has a porous, sponge-like texture. Its crystalline structure consists of stacked, layered sheets. It does not compress when wet and has good water absorption properties, capable of holding up to 500% (w/w) of water (Department of Mineral Resources, 2008). When used as a pelleting material, vermiculite must be finely ground to a size of 1–4 μm to form pellets effectively. The surface of the pellets is smooth and uniform (Singsopa *et al.*, 2024). Meanwhile, bentonite and pumice were similarly easy. Pumice and bentonite have porous, sponge-like particles that absorb moisture well and can combine with binders effectively. However, they are not very easy to pellet because they set quickly and form solid pellets rapidly. Notably, talcum presented

moderate difficulty for pelleting (Table 1). Although talcum is a fine powder that absorbs moisture well, the hardness of only 1 Mohs and has a slippery surface. Therefore, making talcum difficult for pellets to stick together, as also reported in the pelleting of marigold seeds (Singsopa *et al.*, 2024).

The effectiveness of vermiculite and bentonite as pelleting materials has also been associated with their high cation exchange capacity and favorable moisture retention properties, which improve pellet stability and handling during sowing operations (Pedrini *et al.*, 2017). Furthermore, seed pelleting materials with porous structures have been reported to improve seed-soil contact and maintain a more stable hydration environment during early imbibition (Rocha *et al.*, 2019).

Seed Weight, Size and Dissolution Period

Seed enhancement of small maize seeds (16/64 inch) using four pelleting materials—bentonite, vermiculite, talcum, and pumice—resulted in a consistent increase in seed size to approximately

18/64 inch (S18). Pelleted seeds exhibited a 100-seed weight ranging from 24.65 to 25.32 g (Figure 1), which was substantially higher than that of unpelleted small seeds (S16; 18.73 g), representing an increase of approximately 32%. Seed diameter of pelleted treatments ranged from 9.54 to 9.84 mm, compared with 8.65 mm for unpelleted S16 seeds, making pelleted seeds comparable in size to unpelleted medium- and large-sized seeds (S18 and S20; 9.70–9.78 mm). The increase in seed size and mass improves seed flowability and uniformity, which are critical for precision and mechanical planting systems (Taylor, 2020; Afzal, 2023).

Recent researches have demonstrated that increasing seed uniformity through pelleting can significantly improve singulation efficiency and planting precision under mechanical sowing systems, especially in small-seeded crops and irregularly shaped seeds. Similar improvements are increased sowing accuracy and reduced seed loss during planting (Bozdar *et al.*, 2026).

Table 1 Weight of 100 seeds, seed size, and dissolution of pelleted maize seeds

Treatment	Difficulty formation	Weight of 100 seeds (g)	Seed size (mm)	Seed dissolution (s)
Seed size #16 + bentonite	4	24.69 ± 0.80 ^b	9.63 ± 0.09 ^a	0.38 ± 0.04 ^a
Seed size #16 + vermiculite	5	24.65 ± 0.83 ^b	9.54 ± 0.26 ^a	0.40 ± 0.04 ^a
Seed size #16 + talcum	3	25.23 ± 0.16 ^b	9.56 ± 0.34 ^a	0.41 ± 0.06 ^a
Seed size #16 + pumice	4	25.32 ± 0.18 ^b	9.84 ± 0.15 ^a	0.24 ± 0.03 ^b
Seed size #16	-	18.73 ± 0.37 ^c	8.65 ± 0.12 ^b	0.00 ^c
Seed size #18	-	24.87 ± 0.87 ^b	9.70 ± 0.18 ^a	0.00 ^c
Seed size #20	-	28.60 ± 0.84 ^a	9.78 ± 0.27 ^a	0.00 ^c
P-value		< 0.001	< 0.001	< 0.001
CV (%)	-	2.8	2.3	11.8

Note: Means within the same column followed by different superscript letters (a, b, c) are significantly different ($P < 0.05$) according to Duncan's multiple range test. Values are presented as mean ± standard deviation. Seed size #16, #18, and #20 mean 16/64, 18/64, and 20/64 inch, respectively. The difficulty of formation for seed pelleting: 1 = very difficult, 2 = difficult, 3 = moderate, 4 = easy, and 5 = very easy. CV = coefficient of variation.

The dissolution period of pelleted seeds differed significantly among pelleting materials (Table 1). Seeds pelleted with bentonite, vermiculite, and talcum required approximately 0.38–0.41 s for initial coating disintegration, whereas pumice-pelleted seeds dissolved more rapidly (0.24 s). These differences can be attributed to the physical structure, porosity, and water-holding capacity of the pelleting materials.

Bentonite and vermiculite are characterized by high water-retention capacity and swelling properties,

enabling pellets to absorb and retain moisture around the seed surface after sowing (Grellier *et al.*, 1999; Saenghin and Siri, 2014). This temporary moisture reservoir helps maintain a favorable hydration gradient during early imbibition, thereby reducing rapid water loss under fluctuating soil moisture conditions. In addition, the layered structure of vermiculite promotes controlled water uptake and gradual pellet disintegration, which may prevent excessive flooding of the seed and enhance metabolic reactivation during germination

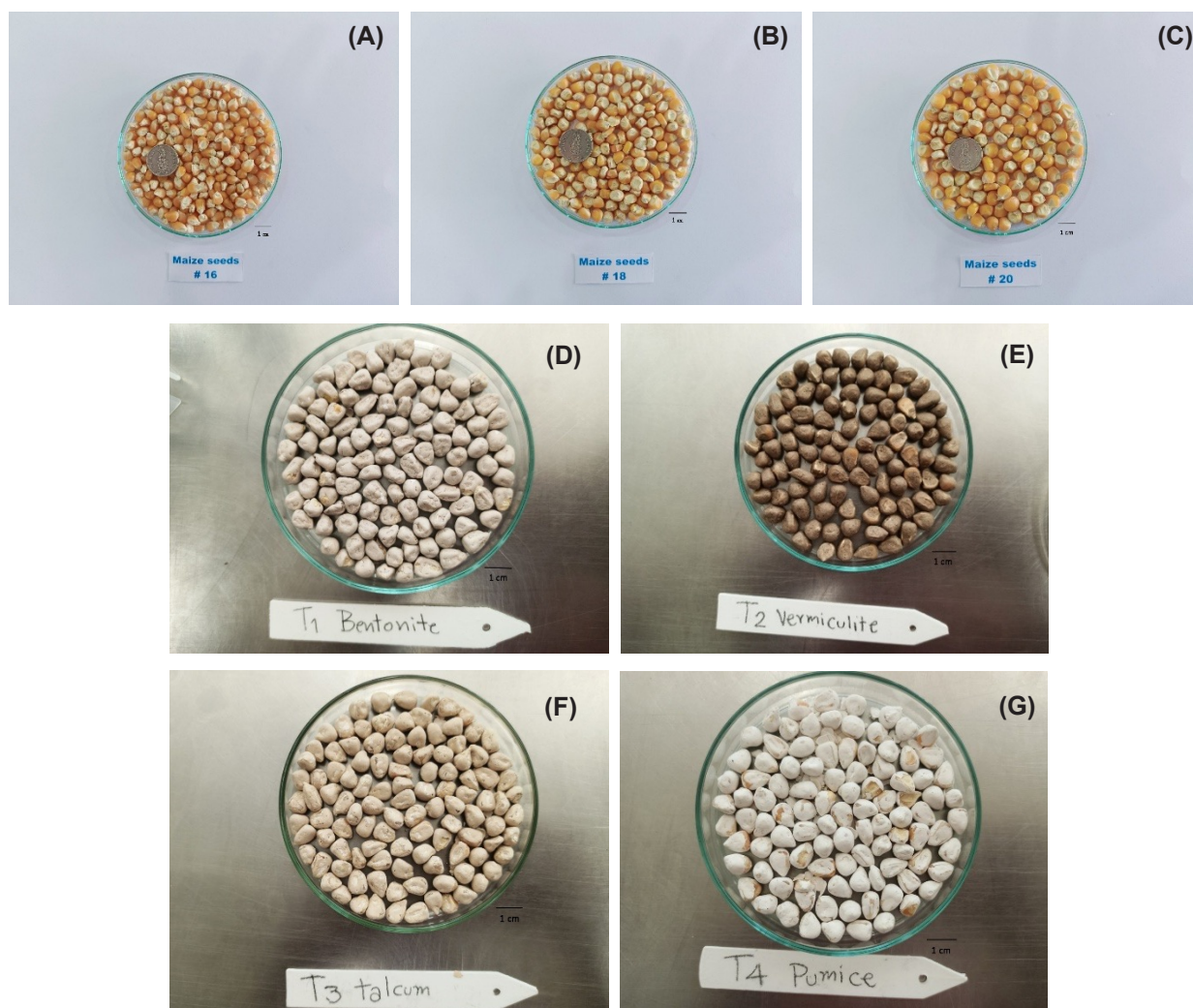


Figure 1 Physical characteristics of maize seeds before and after pelleting. A–C show unpelleted maize seeds of size classes #16 (16/64 inch), #18 (18/64 inch), and #20 (20/64 inch), respectively. D–G show small-sized maize seeds (#16) after pelleting with bentonite, vermiculite, talcum, and pumice, respectively. All images were captured at the same magnification. Scale bar = 5 mm.

(Taylor *et al.*, 1998). Bentonite, with its high water-holding capacity of approximately 173–184%, has also been associated with rapid imbibition and faster completion of the lag phase, leading to improved germination performance (Nuyttens *et al.*, 2013).

Oxygen diffusion is another critical factor influencing early germination and seedling vigor. Pellets with moderate dissolution rates, such as those formed with bentonite and vermiculite, tend to maintain adequate pore spaces, allowing sufficient oxygen diffusion while retaining moisture (Scott, 1989; Halmer, 2000). Excessively rapid dissolution, as observed with pumice, may lead to the quick collapse of the pellet structure, reducing its ability to regulate water availability around the seed. Conversely, overly slow dissolution could restrict oxygen diffusion and delay radicle protrusion; however, the dissolution times observed in this study remained within an optimal range for maize germination.

Pumice, which has a highly porous but rigid structure, dissolves more rapidly in water and exhibits lower water-holding capacity compared with bentonite and vermiculite. Although this characteristic promotes fast water penetration, it may reduce the duration of moisture retention around the seed, particularly under field conditions where evaporation and soil heterogeneity are significant (Grellier *et al.*, 1999).

Similar findings have been reported in pelleted vegetable seeds, where vermiculite-based pellets showed slower dissolution and improved seedling growth due to enhanced moisture retention and aeration balance (Kangsopa *et al.*, 2021). Overall, the results indicate that pelleting materials with moderate dissolution rates, high water retention capacity, and adequate porosity, such as bentonite and vermiculite, are more effective in improving the physical quality of small maize seeds and creating a favorable microenvironment for early germination and seedling establishment.

Seed Germination in Seed Laboratory Testing

Pelleting of small maize seeds (S16) with bentonite, vermiculite, talcum, or pumice did not significantly affect final germination percentage compared with unpelleted S16 seeds (Table 2). Germination of pelleted seeds ranged from 87.00 to 89.00%, which was statistically similar to that of unpelleted S16 seeds (93.50%). These results indicate that the pelleting materials did not impede water uptake, gas exchange, or radicle emergence during germination. In contrast, unpelleted medium- and large-sized seeds (S18 and S20) exhibited significantly greater germination percentages (97.00–98.50%) than S16 seeds, reflecting the well-documented advantage of larger seeds possessing greater nutrient reserves and higher seed vigor (TeKrony and Egli, 1991; Finch-Savage and Bassel, 2016).

The present findings are consistent with previous reports demonstrating that properly formulated seed coatings do not adversely affect germination when pellet composition and disintegration characteristics are optimized (Pedrini *et al.*, 2017; Sharma *et al.*, 2021). Similarly, Pipiak *et al.* (2024) reported that bentonite-based pelleting materials can enhance maize seedling growth and subsequent crop performance without compromising germination. The higher germination observed in larger seed classes further supports the importance of seed vigor for successful crop establishment, as vigorous seeds generally promote rapid emergence, uniform stand establishment, and improved yield potential.

The maintenance of high germination percentages in pelleted seeds suggests that pellet disintegration occurred rapidly enough to permit normal imbibition and oxygen diffusion. This observation agrees with previous studies indicating that effective pelleting materials should readily disintegrate after sowing to avoid restricting germination processes

(Taylor *et al.*, 1998; Halmer, 2000). In the present study, pellet dissolution times ranged from 0.24 to 0.41 s, indicating rapid disintegration and minimal resistance to germination. These results are consistent with earlier studies reporting that bentonite- and vermiculite-based pelleting materials effectively preserve germination in maize and sweet maize seeds (Woraphandhamakul and Siri, 2011; Sakhamula and Siri, 2012).

Speed of Germination in Seed Laboratory Testing

Despite comparable final germination percentages, pelleted seeds exhibited a slower speed of germination (8.20–9.04 plants/day) than unpelleted S18 (10.84 plants/day) and S20 seeds (9.78 plants/day; Table 2). This delay is consistent with the presence of a coating layer, which can temporarily moderate water uptake and oxygen diffusion during the early imbibition phase (Scott, 1989; Taylor *et al.*, 1998). Larger unpelleted seeds generally exhibit faster germination due to higher metabolic activity and greater reserve availability, supporting more rapid radicle protrusion (Egli and Rucker, 2012).

Among pelleted treatments, vermiculite- and talcum-pelleted seeds exhibited germination speeds similar to S16 seeds, indicating that these materials allowed relatively rapid water penetration. Vermiculite has a layered structure and moderate porosity, facilitating water absorption while maintaining internal aeration (Halmer, 2000). Talcum-based pellets, although uniform and smooth, may form denser coatings that slightly delay imbibition, explaining the absence of improvement in germination speed relative to unpelleted S16 seeds.

Seed Vigor Assessed by Accelerated Aging (AA test) in Seed Laboratory Testing

Accelerated aging clearly differentiated the effects of pelleting materials on seed vigor (Table 2).

Seeds pelleted with bentonite and vermiculite maintained significantly higher post-aging germination (64.50–67.00%) than unpelleted S16 seeds (47.00%) and were statistically comparable to S18 seeds (63.25%). These results indicate that pelleting with bentonite or vermiculite enhanced resistance to physiological deterioration.

The superior AA test performance of these materials can be attributed to their high water-holding capacity and buffering properties, which help regulate moisture uptake during re-imbibition after aging stress. Controlled hydration reduces membrane leakage and metabolic disruption, key determinants of seed vigor following exposure to high temperature and humidity (TeKrony, 2003; Bewley *et al.*, 2013). Vermiculite and bentonite pellets likely moderated rapid water influx, thereby preserving membrane integrity during the lag phase of germination.

The superior performance of bentonite and vermiculite pelleted seeds may be attributed to their ability to regulate water uptake during the early phases of imbibition. Rapid water uptake can cause membrane disruption and solute leakage, particularly in aged or small seeds with limited reserve tissues (Walters *et al.*, 2010; Bewley *et al.*, 2013). The high water-holding capacity of bentonite and vermiculite likely buffered moisture fluctuations around the seed, resulting in more gradual hydration and improved membrane stability. This mechanism may explain the higher post-aging germination observed in the AA test.

In addition to water relations, oxygen availability within the pellet matrix may have influenced germination and seedling establishment. Pelleting materials with moderate porosity can simultaneously retain moisture and maintain gaseous diffusion pathways, thereby supporting aerobic respiration during germination (Taylor *et al.*, 1998; Halmer, 2000). Vermiculite, owing to its layered structure and high porosity, may provide a more favorable balance between water retention

and oxygen diffusion than denser pelleting materials. Similar observations have been reported in vegetable seed pelleting systems, where vermiculite-based coatings improved seedling establishment under suboptimal moisture conditions (Pedrini *et al.*, 2017).

In contrast, seeds pelleted with talcum and pumice showed post-aging germination below 60%, consistent with earlier reports indicating lower protection against vigor loss (Saenghin and Siri, 2014). Pumice, despite its high porosity, has relatively low water retention capacity, which may limit its ability to buffer moisture stress during aging.

The present findings are generally consistent with previous studies reporting beneficial effects of bentonite and vermiculite on seed vigor and seedling establishment (Sakhamula and Siri, 2012; Pedrini *et al.*, 2017). However, unlike some reports that observed significant improvements in final germination following pelleting, the current study found no increase in germination percentage relative to unpelleted seeds. This discrepancy may be attributed to differences in seed species, seed lot quality, coating thickness, and environmental conditions during germination. In the present study, the primary benefit of pelleting appeared to be improved seed vigor and stress tolerance rather than enhancement of final germination capacity. These results suggest that the physiological advantages of pelleting are more closely associated with improved hydration dynamics and protection against deterioration than with direct stimulation of germination processes.

Seedling Growth under Laboratory Conditions

Seedling growth responses further reflected differences among pelleting materials. At 7 days after sowing, talcum-pelleted seeds produced the longest shoots (12.02 cm) and roots (16.01 cm), exceeding those of other pelleted treatments and unpelleted seeds (Table 2). This enhanced early growth suggests that talcum pellets allowed rapid initial hydration,

supporting early cell expansion once germination had occurred.

Talcum is widely used in seed pelleting due to its fine particle size and ability to form uniform coatings, which can promote even water distribution around the seed (Afzal *et al.*, 2020). However, the strong early growth response observed with talcum did not correspond to improved performance in the AA test, indicating that early seedling growth under favorable conditions does not necessarily reflect higher seed vigor or storability. This distinction highlights the importance of evaluating multiple seed quality parameters when assessing pelleting effects (Finch-Savage and Bassel, 2016).

Seed Germination under Field Condition

Under field conditions, final germination of pelleted seeds ranged from 93.00 to 95.50% and did not differ significantly from unpelleted seeds of all size classes (S16, S18, and S20; 91.00–94.00%; Table 3). This confirms the laboratory results, indicating that pelleting did not restrict germination under either controlled or natural conditions. The consistently high germination across treatments suggests that pellet dissolution occurred rapidly enough to permit adequate water uptake and oxygen diffusion, which are critical for maize seed germination (Taylor *et al.*, 1998; Halmer, 2000).

Seeds pelleted with bentonite exhibited the highest germination (95.50%), although differences among treatments were not statistically significant. Bentonite pellets have high water absorption capacity and moderate hardness, which can promote uniform seed hydration during the lag phase without imposing mechanical resistance to radicle emergence (Javed and Afzal, 2018). Importantly, the absence of germination reduction in pelleted seeds demonstrates that the physical modification of small seeds did not negatively affect their germination potential.

Speed of Germination under Field Condition

Pelleted seeds exhibited significantly higher germination speed (11.02–11.84 plants/day) than unpelleted S16, S18, and S20 seeds (8.16–8.52 plants/day; Table 3). This trend contrasts with laboratory results, where pelleted seeds germinated more slowly, and highlights the importance of field conditions in evaluating seed vigor. In the field, fluctuating moisture and temperature conditions favor pelleted seeds with greater water-holding capacity, as the pellet can act as a temporary moisture buffer around the seed. Bentonite and vermiculite pellets, in particular, can absorb and retain water during intermittent drying periods, thereby maintaining seed hydration and promoting faster radicle emergence (Scott, 1989; Finch-Savage and Bassel, 2016). Consequently, pelleted seeds exhibited more rapid and uniform emergence despite having final germination percentages similar to those of unpelleted seeds.

Accelerated Aging Test under Field Condition

After accelerated aging, field germination of all pelleted treatments remained above 66%, with bentonite-pelleted seeds showing the highest germination (70.82%), followed by pumice (67.81%), talcum (67.80%), and vermiculite (66.09%; Table 3). In contrast, unpelleted seeds of all size classes (S16, S18, and S20) exhibited substantially lower post-aging germination (49.24–51.00%), indicating that pelleting improved seed vigor under aging stress.

Across both laboratory and field evaluations, pelleted S16 seeds consistently outperformed unpelleted S16 seeds after AA treatment (Tables 2–3). This trend reflects the greater susceptibility of small seeds to aging-induced deterioration and imbibitional injury due to limited reserve tissues and reduced structural protection (TeKrony and Egli, 1991; McDonald, 1999; TeKrony, 2003; Bewley *et al.*, 2013).

Pelleting with bentonite and vermiculite likely

reduced post-aging injury by moderating water uptake and preserving membrane integrity. Pellets with moderate hardness and high water-holding capacity slow initial imbibition, reducing membrane rupture during rehydration (Taylor *et al.*, 1998; Halmer, 2000). Furthermore, the moisture-buffering properties of bentonite and vermiculite promote gradual hydration of aged seeds, limiting solute leakage and vigor loss under stress (Walters *et al.*, 2010; Nuyttens *et al.*, 2013; Finch-Savage and Bassel, 2016).

In contrast, talcum- and pumice-pelleted seeds exhibited slightly lower post-aging germination, likely due to lower water retention and reduced buffering against rapid re-imbibition, consistent with previous studies of pelleted maize and vegetable seeds (Saenghin and Siri, 2014). Overall, the AA test results indicate that pelleting—particularly with bentonite and vermiculite—enhances stress tolerance and physiological quality of small maize seeds under field conditions by mitigating aging-related membrane damage and imbibitional injury.

Seedling Growth under Field Conditions

Pelleted seeds generally promoted maize seedling growth compared with unpelleted seeds. Pelleted seeds produced longer shoots (15.24–16.74 cm) than unpelleted seeds of all three seed sizes (9.27–9.67 cm). Shoot length was similar among the pelleting materials. Root length was greatest with bentonite (14.14 cm) and vermiculite (14.06 cm), followed by talcum (13.33 cm), whereas pumice resulted in shorter roots (11.07 cm). Unpelleted seeds had root lengths ranging from 10.79 to 11.18 cm across seed sizes (Table 3). Overall, pelleting had a greater effect on shoot length than on root length under field conditions.

The porous structure of vermiculite facilitates rapid water imbibition and adequate oxygen diffusion during germination. Furthermore, its negatively charged

Table 2 Effect of pelleting on maize seed quality in the seed laboratory testing

Treatment	Seed laboratory testing				
	Germination (%)	Speed of germination (plant/day)	AA test (%)	Shoot length (cm)	Root length (cm)
Seed size #16 + bentonite	87.00 ± 2.58 ^b	8.20 ± 0.38 ^{cd}	67.00 ± 2.58 ^a	10.78 ± 0.17 ^{bc}	14.58 ± 0.55 ^{ab}
Seed size #16 + vermiculite	89.00 ± 2.64 ^b	8.56 ± 0.74 ^{cd}	64.50 ± 1.98 ^a	10.37 ± 0.38 ^{cde}	13.61 ± 0.89 ^{bc}
Seed size #16 + talcum	87.50 ± 3.72 ^b	8.32 ± 0.98 ^{cd}	55.50 ± 1.29 ^b	12.02 ± 0.15 ^a	16.01 ± 1.09 ^a
Seed size #16 + pumice	87.50 ± 3.41 ^b	9.04 ± 0.15 ^{bc}	53.25 ± 2.77 ^b	11.04 ± 0.18 ^b	14.70 ± 1.14 ^{ab}
Seed size #16	93.50 ± 2.51 ^b	8.60 ± 0.14 ^b	47.00 ± 3.16 ^c	10.57 ± 0.40 ^{cd}	15.30 ± 0.49 ^a
Seed size #18	98.50 ± 1.90 ^a	10.84 ± 0.30 ^a	63.25 ± 2.77 ^a	10.23 ± 0.37 ^{de}	13.78 ± 0.83 ^{bc}
Seed size #20	97.00 ± 3.46 ^a	9.78 ± 0.43 ^b	57.25 ± 2.75 ^b	10.17 ± 0.06 ^e	14.55 ± 1.19 ^{ab}
P-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
CV (%)	6.0	6.3	5.2	2.6	6.4

Note: Means within the same column followed by different superscript letters (a, b, c, d, e) are significantly different ($P < 0.05$) according to Duncan's multiple range test. Values are presented as mean ± standard deviation. Seed size #16, #18, and #20 mean 16/64, 18/64, and 20/64 inch, respectively. CV = coefficient of variation.

Table 3 Effect of pelleting on maize seed quality in the field condition

Treatment	Field condition				
	Germination (%)	Speed of germination (plant/day)	AA test (%)	Shoot length (cm)	Root length (cm)
Seed size #16 + bentonite	95.50 ± 1.91	11.84 ± 0.24 ^a	70.82 ± 0.27 ^a	16.74 ± 1.90 ^a	14.14 ± 0.44 ^{ab}
Seed size #16 + vermiculite	94.50 ± 2.51	11.02 ± 0.45 ^b	66.09 ± 0.11 ^b	16.48 ± 1.53 ^a	14.06 ± 0.43 ^a
Seed size #16 + talcum	93.00 ± 1.83	11.30 ± 0.51 ^{ab}	67.80 ± 0.19 ^{ab}	15.40 ± 0.73 ^a	13.33 ± 0.21 ^b
Seed size #16 + pumice	94.50 ± 1.91	11.28 ± 0.59 ^{ab}	67.81 ± 0.24 ^{ab}	15.24 ± 1.16 ^a	11.07 ± 0.55 ^c
Seed size #16	91.00 ± 2.03	8.16 ± 0.44 ^c	49.24 ± 0.30 ^c	9.27 ± 0.27 ^b	10.79 ± 0.89 ^c
Seed size #18	94.00 ± 2.32	8.38 ± 0.26 ^c	49.86 ± 0.16 ^c	9.46 ± 0.47 ^b	11.18 ± 0.52 ^c
Seed size #20	94.00 ± 2.58	8.52 ± 0.51 ^c	51.00 ± 0.21 ^c	9.67 ± 0.73 ^b	10.91 ± 0.21 ^c
P-value	0.70	< 0.001	< 0.001	< 0.001	< 0.001
CV (%)	3.9	4.5	4.8	8.5	5.6

Note: Means within the same column followed by different superscript letters (a, b, c) are significantly different ($P < 0.05$) according to Duncan's multiple range test. Values are presented as mean ± standard deviation. Seed size #16, #18, and #20 mean 16/64, 18/64, and 20/64 inch, respectively. CV = coefficient of variation.

surface can adsorb cationic compounds, thereby reduce potential phytotoxic effects and promote early seedling growth and establishment (Reis *et al.*, 2025).

In addition, bentonite possesses high water-holding and cation exchange capacities, which may contribute to improved moisture retention and nutrient availability around the seed, thereby promoting faster germination and improved seedling establishment (Noble *et al.*, 2001).

In field conditions, fluctuating soil moisture and temperature can influence seedling establishment, particularly in small-sized maize seeds. Pelleted seeds with high water-holding capacity may provide an advantage because the pellet functions as a temporary moisture reservoir around the seed. Bentonite and vermiculite, in particular, can absorb and retain water during intermittent drying periods, thereby maintaining seed hydration and promoting more rapid radicle emergence (Scott, 1989; Finch-Savage and Bassel, 2016). Consequently, pelleted seeds exhibited greater seedling vigor and more uniform field emergence, despite showing similar final germination percentages to unpelleted seeds.

CONCLUSIONS

Pelleting small maize seeds with bentonite or vermiculite effectively increased seed size and weight, improved flowability, and maintained high germination under laboratory and field conditions. These materials enhanced post-aging vigor by moderating water uptake, buffering moisture during

early imbibition, and providing physical protection to the embryo, thereby reducing membrane damage and imbibitional injury, particularly in small seeds with limited reserves. Talcum and pumice, while promoting early seedling growth, offered less protection against aging stress due to lower water retention and moisture-buffering capacity. Overall, bentonite and vermiculite are superior pelleting materials for improving both the physiological quality and field performance of small maize seeds, demonstrating their value for enhancing uniform germination, and seedling establishment.

CONFLICT OF INTEREST

The authors declare that they have no financial or personal conflicts of interest, nor any relationships or activities related to seed use or research conduct that could have influenced the results presented in this publication. All authors have given their consent for publication.

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